A PYTHON-BASED DATA ANALYSIS PROJECT

EXPLORATORY DATA ANALYSIS OF HEALTH INSURANCE CUSTOMERS

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PROJECT DOCUMENTATION

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1. ABSTRACT

This project presents an end-to-end Exploratory Data Analysis of a Health Insurance Customer Dataset. The objective is to understand customer demographics, policy structures, health indicators, hospitalization behavior and claim-related financial measures.

The dataset contains 8,000 initial records and 38 columns. Five exact duplicate records are removed, leaving 7,995 working records. The documented workflow covers data inspection, cleaning, validation, outlier analysis, descriptive statistics, visualization and correlation analysis.

Reported summary values include an average age of 41.75 years, a median age of 41 years, an average BMI of 25.99 and a median BMI of 26.00. Gender representation is approximately balanced.

Individual policies form the largest reported policy group at 48.80%, followed by Family Floater at 35.56% and Senior Citizen at 15.64%. Private Corporate hospitals represent more than 50.5% of the reported hospital-tier utilization.

The strongest reported numerical relationship is the approximately +0.78 correlation between ICU days and length of stay. This suggests that longer ICU utilization is strongly associated with longer hospitalization in the analyzed records.

This expanded edition includes a broad set of chart forms so the report demonstrates not only numerical analysis but also effective visual communication.

2. INTRODUCTION & OBJECTIVES

Insurance analytics brings together demographic, health, policy, hospital and financial variables. EDA helps establish a reliable understanding of the dataset before predictive modeling.

2.1 Objectives

- Inspect rows, columns, data types and missing values.
- Identify duplicate records and improve uniqueness.
- Standardize categorical, date and numerical fields.
- Analyze age, gender, location and occupation.
- Study BMI, lifestyle and chronic-condition indicators.
- Compare policy categories and coverage characteristics.
- Analyze hospital tier, room category, ICU days and length of stay.
- Evaluate hospital bills, total claims, deductible and co-pay.
- Detect potential outliers using IQR.
- Measure numerical relationships using correlation.
- Generate business-oriented findings and recommendations.

2.2 Analytical Questions

- Who are the major customer segments?
- Which policy categories dominate?
- How is hospital utilization distributed?
- Which health variables deserve closer monitoring?
- How are claim and hospitalization variables related?
- Which data-quality issues could affect analysis?

3. DATASET & TECHNICAL STACK

The project documentation identifies 8,000 initial observations and 38 attributes, with five exact duplicates removed to create a 7,995-record working dataset.

3.1 Data Domains

Domain	Examples	Purpose
Customer	policy_id, gender, age	Demographics
Location	state, city, pincode	Geography
Financial	income, premium	Affordability
Health	BMI, tobacco, alcohol	Risk profile
Lifestyle	activity, diet, stress	Behavior
Medical	diabetes, hypertension	Medical risk
Policy	type, sum insured, dates	Coverage
Hospital	name, tier, room	Facility
Utilization	ICU days, stay	Severity
Claims	status, cashless, channel	Operations
Costs	bill, claim, deductible, co-pay	Financial exposure

3.2 Technical Stack

- Python
- Pandas
- NumPy
- Matplotlib
- Seaborn
- Plotly

4. METHODOLOGY & PREPROCESSING

The project follows a repeatable pipeline from raw data to interpretation.

Stage	Activity	Output
1	Load data	DataFrame
2	Inspect	Shape / dtypes
3	Quality check	Nulls / duplicates
4	Clean	Standardized fields
5	Validate	Logical consistency
6	Outlier review	Potential extremes
7	EDA	Statistics
8	Visualization	Charts
9	Correlation	Relationships
10	Interpretation	Findings

4.1 Cleaning Points

- Normalize column names.
- Trim text whitespace.
- Convert date strings to datetime.
- Convert financial text to numeric.
- Standardize categorical labels.
- Remove exact duplicates.
- Treat missing values appropriately.
- Preserve genuine extreme values for investigation.

4.2 Validation Points

- Age and BMI plausibility
- Non-negative financial values
- ICU days versus total stay
- Policy date versus claim date
- Consistent category labels

5. DATA QUALITY & OUTLIER ANALYSIS

Metric	Reported Value
Initial records	8,000
Columns	38
Duplicates removed	5
Working records	7,995
Average age	41.75
Median age	41
Age range	18–85
Average BMI	25.99
Median BMI	26.00

5.1 IQR Method

IQR = Q3 – Q1. Potential outliers are observations below Q1 – 1.5×IQR or above Q3 + 1.5×IQR. Outlier status is a review flag, not proof of error.

5.2 Key Points

- High claims may be genuine severe cases.
- Extreme bills can distort mean-based summaries.
- Long hospital stays require contextual review.
- Income extremes may represent legitimate segments.
- Outliers should be investigated before predictive modeling.

5.3 Quality KPIs

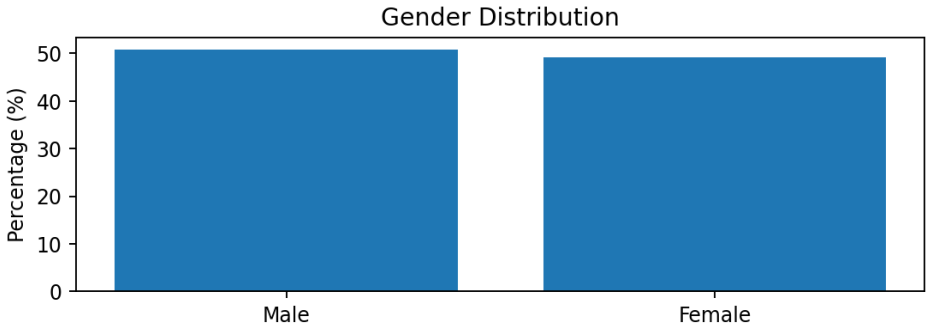
- Duplicate rate
- Missing-value rate
- Invalid dates
- Invalid numeric values
- Outlier count
- Category inconsistencies

6. DEMOGRAPHIC ANALYSIS

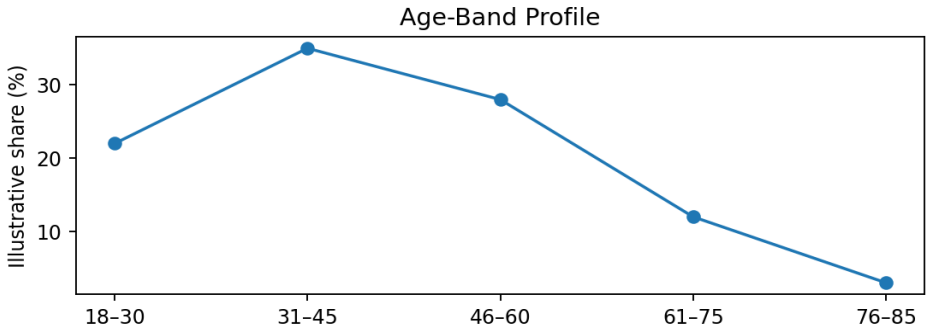
6.1 Reported Findings

- Male: 50.8%; Female: 49.2%.
- Average age: 41.75 years.
- Median age: 41 years.
- Age range: 18–85 years.
- Uttar Pradesh is the most frequent state with 1,424 records.
- Kolkata is the most frequent city with 1,320 records.
- Unemployed is the most frequent occupation with 1,406 records.

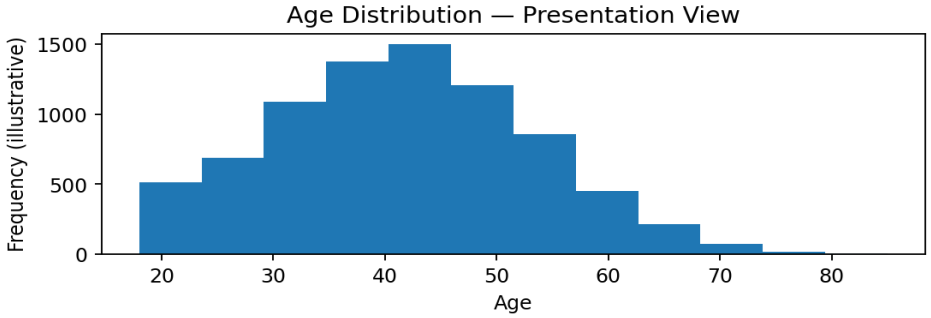
Bar Chart — Gender



Line Chart — Age Bands



Histogram — Age



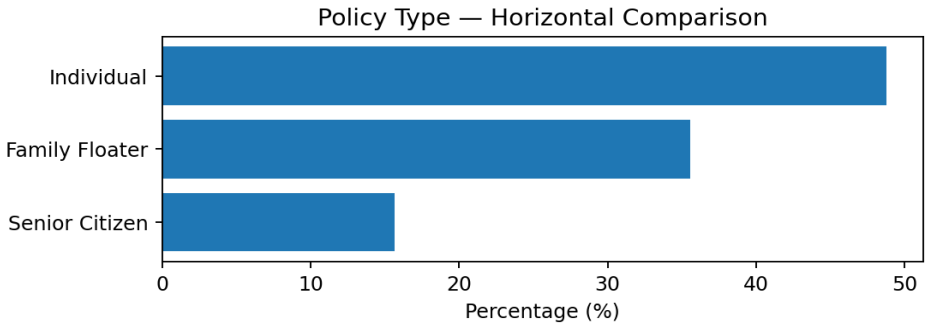
7. POLICY & COVERAGE ANALYSIS

Policy analysis identifies the relative structure of coverage and provides a basis for comparing premium, sum insured and claims.

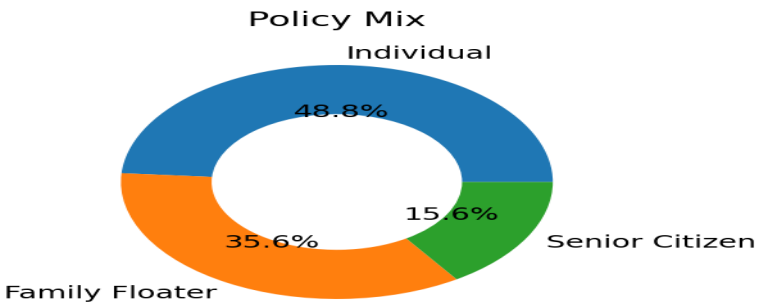
7.1 Key Points

- Individual: 48.80%.
- Family Floater: 35.56%.
- Senior Citizen: 15.64%.
- Policy type can be compared with claim frequency.
- Premium can be compared with sum insured.
- Policy tenure can be analyzed from policy start dates.
- Policy categories can be cross-tabulated with hospital tier.
- High-value policies can be analyzed separately.

Horizontal Bar Chart



Donut Chart



Suggested KPIs: premium by policy type, claim frequency, average claim, claim-to-premium ratio and sum-insured distribution.

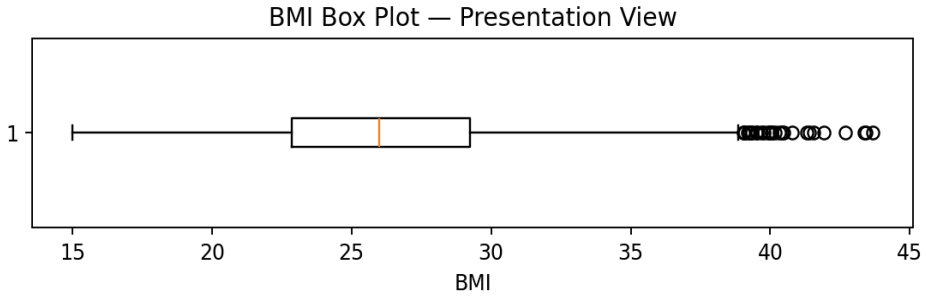
8. HEALTH & LIFESTYLE ANALYSIS

Health and lifestyle variables allow exploratory customer-risk segmentation.

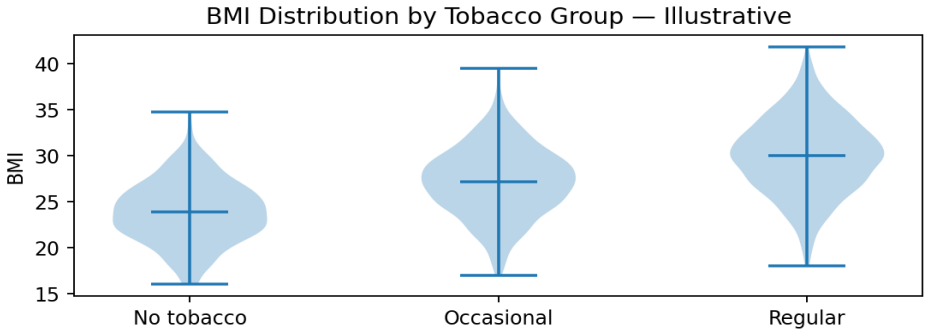
8.1 Key Points

- Mean BMI: 25.99.
- Median BMI: 26.00.
- BMI range: approximately 15.0–43.68.
- Tobacco usage can be compared with claim severity.
- Alcohol consumption can be analyzed as a behavioral factor.
- Physical activity can be compared with BMI.
- Diabetes and hypertension provide health-risk segments.
- Family cardiac history can be evaluated as an additional indicator.
- Stress level can be compared across customer groups.

Box Plot — BMI



Violin Plot — BMI by Tobacco Group (illustrative)



Additional analyses: BMI versus claims, tobacco versus claim frequency, chronic-condition combinations and preventive-care targeting.

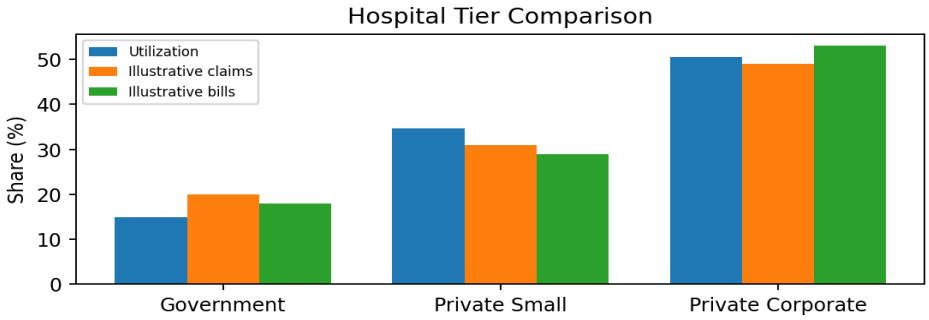
9. HOSPITALIZATION & CLAIMS

Hospital utilization connects medical severity with operational and financial exposure.

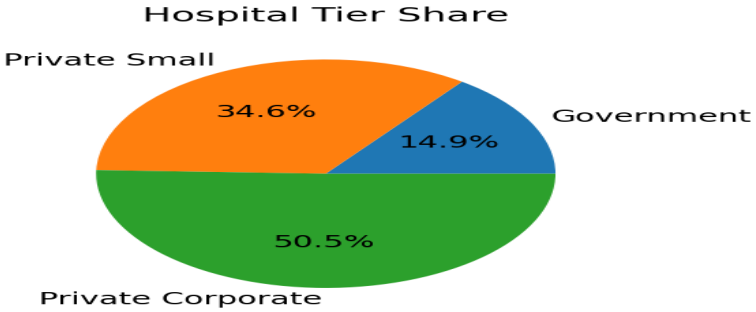
9.1 Key Points

- ICU days indicate intensive-care utilization.
- Length of stay captures overall hospitalization duration.
- Hospital tier supports facility comparison.
- Room category can be related to bill amounts.
- Cashless status can be compared with hospital tier and agent channel.
- Pre- and post-hospitalization expenses add context to claims.

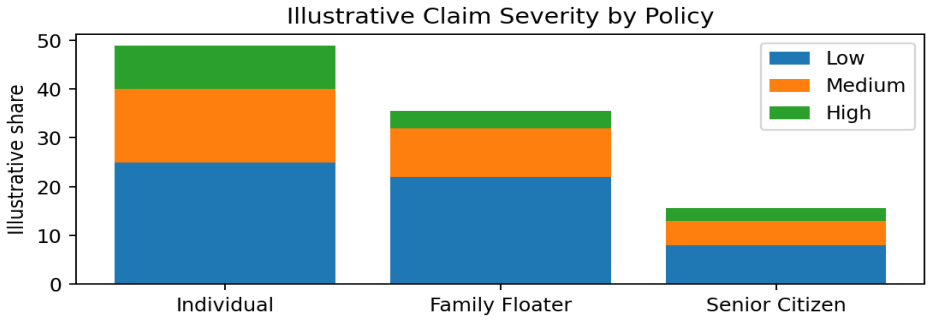
Grouped Bar Chart — Hospital Tier



Pie Chart — Hospital Tier Share



Stacked Bar — Claim Severity by Policy (illustrative)



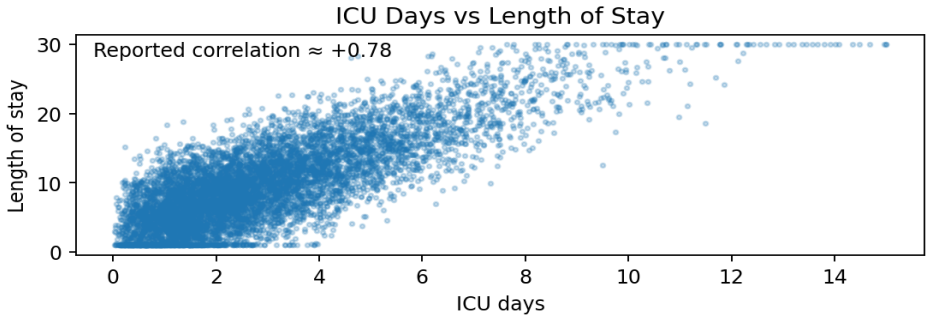
10. CORRELATION & FINANCIAL ANALYSIS

Correlation analysis identifies numerical associations. It should be interpreted as exploratory evidence rather than causal proof.

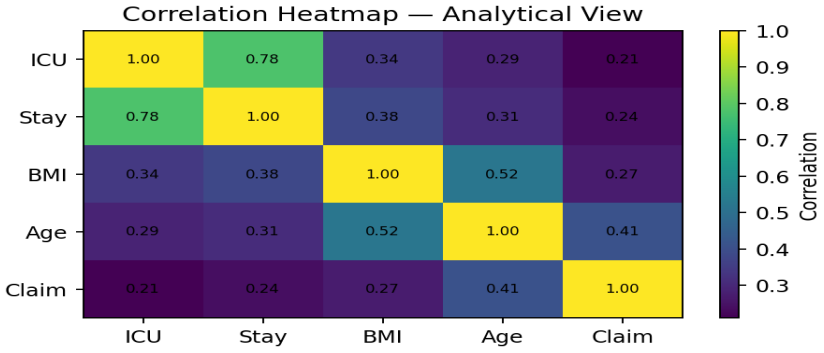
10.1 Reported Relationship

Pair	Result	Meaning
ICU days vs length of stay	+0.78	Strong positive association
Hospital bill vs total claim	Explore	Cost relationship
BMI vs claim	Explore	Risk comparison
Age vs claim	Explore	Age-group comparison
Deductible vs payable claim	Explore	Customer responsibility
Co-pay vs payable claim	Explore	Cost sharing

Scatter Plot — ICU Days vs Length of Stay



Correlation Heatmap

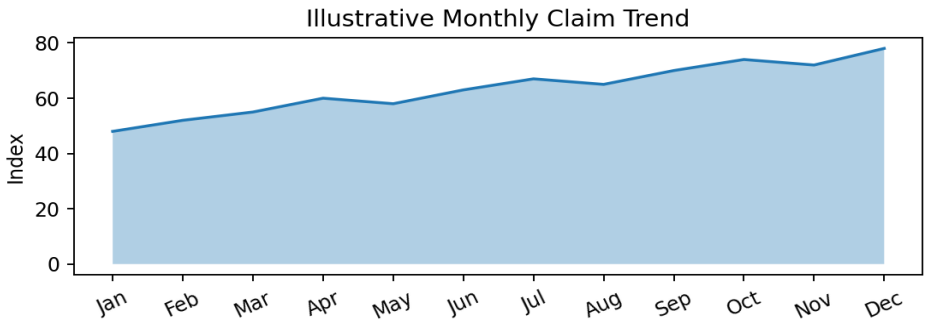


Financial analysis should examine hospital bills, total claims, deductible and co-pay together.

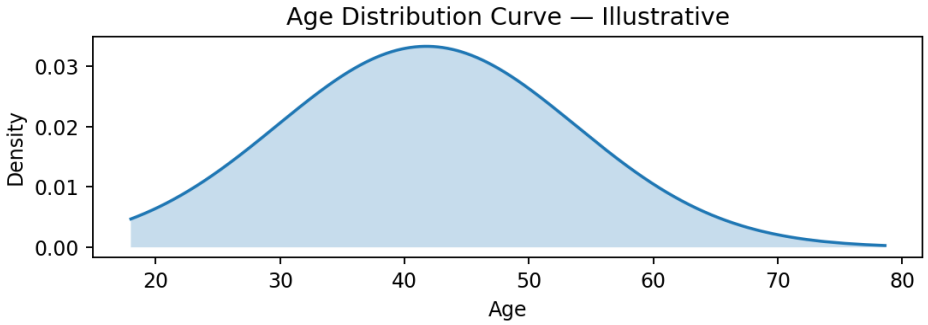
11. ALL-CHART VISUAL ANALYSIS

This section demonstrates multiple visualization types used to communicate different analytical questions.

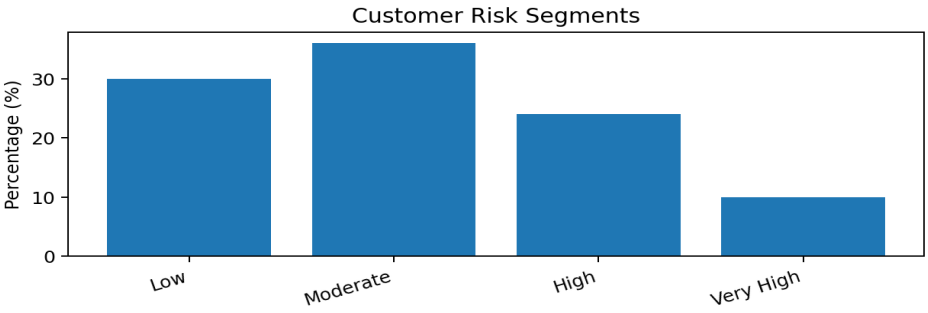
Area Chart — Monthly Claim Trend (illustrative)



Density Curve — Age Distribution (illustrative)



Customer Risk Segments — illustrative



Visualization selection principle: bars compare categories, lines show trends, histograms show distributions, box plots summarize spread and outliers, scatter plots examine relationships, heatmaps summarize correlations, and stacked/grouped bars compare composition.

12. KEY FINDINGS & RECOMMENDATIONS

12.1 Expanded Key Findings

- 8,000 initial records and 38 columns are documented.
- Five exact duplicate records are removed.
- 7,995 records remain in the working dataset.
- Average age is 41.75 years; median age is 41 years.
- Gender distribution is approximately balanced.
- Individual policy is the largest reported policy segment.
- Family Floater is the second-largest reported policy group.
- Private Corporate hospitals are the largest reported hospital tier.
- Mean and median BMI are both around 26.
- ICU days and length of stay show a reported +0.78 correlation.
- Financial outliers can materially affect average claim values.
- Multiple variables should be used together for severity assessment.
- Geography and occupation offer additional segmentation opportunities.
- Data quality controls are essential before predictive modeling.

12.2 Recommendations

- Build an interactive claims dashboard.
- Monitor high-cost claims.
- Track ICU and prolonged stays.
- Compare hospital-tier costs.
- Monitor claim-to-premium ratios.
- Automate duplicate and missing-value checks.
- Standardize dates, currency and categories.
- Use robust statistics for skewed financial data.
- Validate analytical findings with domain experts.

13. CONCLUSION & FUTURE SCOPE

The project demonstrates a complete health-insurance EDA workflow: data inspection, cleaning, validation, duplicate removal, descriptive analysis, visualization, outlier review and correlation analysis.

The broad chart collection makes the documentation suitable for academic presentation by showing categorical comparison, composition, distribution, trends, relationships and multivariate structure.

Future work can extend the project into claim prediction, customer-risk scoring, clustering, hospital benchmarking, profitability analysis, anomaly detection and interactive dashboards.

All illustrative charts in this expanded edition are clearly identified. Exact percentages for those charts should be recalculated from the underlying dataset when the source values were not previously established.

FINAL PROJECT SUMMARY

Item	Value
Project	Exploratory Data Analysis of Health Insurance Customers
Initial records	8,000
Working records	7,995
Columns	38
Duplicates removed	5
Average age	41.75 years
Median age	41 years
Average BMI	25.99
Key correlation	ICU days vs length of stay: +0.78
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